

BSSE2301 SOFTWARE ENGINEERING MINI PROJECT 2026
Python programming learning course outline modules

Modules	Sub modules	Session /Date	Mentor/ Instructor
MODULE 1: INTRODUCTION TO PROGRAMMING AND PYTHON	Introduction to Programming and Python • What is Programming? • Applications of Python • Installing Python and VS Code • Running Python Scripts • Python Syntax • Variables and Data Types	Monday – 8 June 2026 Morning Session	Geoffrey
	Computational Thinking and Problem Solving • Algorithms and Flowcharts • Pseudocode Basics • Understanding Program Logic • Input and Output Operations • Writing Simple Programs	Monday – 8 June 2026 Afternoon Session	Livingstone
	Operators and Expressions • Arithmetic Operators • Comparison Operators • Logical Operators • Assignment Operators • Type Conversion	Wednesday – 10 June 2026 Morning Session	Geoffrey
	Control Structures • Conditional Statements • if, elif, else • Nested Conditions • Real-world Decision Programs	Wednesday – 10 June 2026 Afternoon Session	Livingstone

	Loops in Python • for Loops • while Loops • Loop Control Statements • Nested Loops	Friday – 12 June 2026 Morning Session	Geoffrey
	Mini Project Session • Number Guessing Game • Calculator Program • Student Grading System	Friday – 12 June 2026 Afternoon Session	Livingstone
Module 2: DATA STRUCTURES AND FUNCTIONS	Lists and Tuples • Creating Lists • List Methods • Indexing and Slicing • Tuples and Immutability	Monday – 15 June 2026 Morning Session	Geoffrey
	Sets and Dictionaries • Set Operations • Dictionary Structures • Key-Value Operations • Nested Dictionaries	Monday – 15 June 2026 Afternoon Session	Livingstone
	Functions in Python • Defining Functions • Parameters and Arguments • Return Statements • Scope of Variables	Wednesday – 17 June 2026 Morning Session	Geoffrey
	Advanced Functions • Lambda Functions • Recursive Functions • Built-in Functions • Function-based Projects	Wednesday – 17 June 2026 Afternoon Session	Livingstone
	Modules and Packages	Friday – 19 June 2026	Geoffrey

	• Importing Modules • Creating Custom Modules • Python Standard Library	Morning Session	
	Practical Data Structure Applications • Contact Management System • Inventory Tracker • Student Records System	Friday – 19 June 2026 Afternoon Session	Livingstone
MODULE 3: OBJECT-ORIENTED PROGRAMMING	Introduction to OOP • Classes and Objects • Attributes and Methods • Constructors	Monday – 22 June 2026 Morning Session	Geoffrey
	Encapsulation and Abstraction • Access Modifiers • Data Hiding • Abstract Classes	Monday – 22 June 2026 Afternoon Session	Livingstone
	Inheritance • Parent and Child Classes • Method Overriding • Multiple Inheritance	Wednesday – 24 June 2026 Morning Session	Geoffrey
	Polymorphism • Method Overloading • Method Overriding • Real-world OOP Models	Wednesday – 24 June 2026 Afternoon Session	Livingstone
	Banking System Simulation • ATM Logic • Account Operations • Transactions	Friday – 26 June 2026 Morning Session	Geoffrey
	OOP Project Development	Friday – 26 June 2026 Afternoon Session	Livingstone

	Library Management System		
	File Handling Basics - Reading Files - Writing Files - Appending Data Working with CSV and JSON - CSV Processing - JSON Parsing - Data Serialization	Monday – 29 June 2026 Morning Session	Geoffrey
	Exception Handling - try and except - finally Block - Custom Exceptions	Monday – 29 June 2026 Afternoon Session	Livingstone
	Logging and Debugging - Debugging Techniques - Logging Systems - Error Tracking	Wednesday – 1 July 2026 Morning Session	Geoffrey
	Automation with Python - File Automation - Folder Management - Scheduled Scripts	Wednesday – 1 July 2026 Afternoon Session	Livingstone
	Web Scraping and Automation - Introduction to BeautifulSoup - Data Extraction - Automated Reports	Friday – 3 July 2026 Morning Session	Geoffrey
MODULE 4: DATA SCIENCE AND MACHINE LEARNING	NumPy Fundamentals - Arrays - Matrix Operations - Mathematical Functions	Friday – 3 July 2026 Afternoon Session	Livingstone

	Pandas for Data Analysis • DataFrames • Data Cleaning • Data Transformation	Monday – 6 July 2026 Morning Session	Geoffrey
	Data Visualization • Matplotlib • Plotting Graphs • Charts and Visualization	Monday – 6 July 2026 Afternoon Session	Livingstone
	Exploratory Data Analysis • Statistical Analysis • Pattern Discovery • Dataset Exploration	Wednesday – 8 July 2026 Morning Session	Geoffrey
	Machine Learning Basics • Introduction to ML • Types of Machine Learning • Model Workflow	Wednesday – 8 June 2026 Afternoon Session	Livingstone
	Scikit-Learn Practical • Regression Models • Classification Models • Model Evaluation	Friday – 10 July 2026 Morning Session	Geoffrey
MODULE 5: WEB DEVELOPMENT AND APIs	Django Web Framework • Installing Django • Routing • Templates	Friday – 10 July 2026 Afternoon Session	Livingstone
	HTML Integration with Django • Forms • User Input • Dynamic Pages		
	Flask Web Framework • Installing Flask • Routing	Monday – 13 July 2026	Geoffrey

	• Templates HTML Integration with Flask • Forms • User Input • Dynamic Pages	Morning Session	
	Databases in Python • SQLite Basics • CRUD Operations • Connecting Flask to Database	Monday – 13 July 2026 Afternoon Session	Livingstone
	Databases in Python with Django • SQLite Basics • CRUD Operations • Connecting Flask to Database User Authentication • Login Systems • Sessions • Password Security	Wednesday – 15 July 2026 Morning Session	Geoffrey
	Databases in Python with Flask • SQLite Basics • CRUD Operations • Connecting Flask to Database User Authentication • Login Systems • Sessions • Password Security	Wednesday – 15 July 2026 Afternoon Session	Livingstone
	APIs and HTTP Requests Django • REST APIs • HTTP Methods	Friday – 17 July 2026 Morning Session	Geoffrey

	• requests Library Building APIs with Django • JSON Responses • API Endpoints • API Testing		
	APIs and HTTP Requests Flask • REST APIs • HTTP Methods • requests Library Building APIs with Flask • JSON Responses • API Endpoints • API Testing	Friday – 17 July 2026 Afternoon Session	Livingstone
	Testing in Python • Unit Testing • Writing Test Cases • Assertions	Monday – 20 July 2026 Morning Session	Geoffrey
	Debugging Applications • Debugging Tools • Breakpoints • Performance Optimization	Monday – 20 July 2026 Afternoon Session	Livingstone
MODULE 6: TESTING, DEBUGGING, AND SOFTWARE ENGINEERING	Git and GitHub • Version Control • Repository Management • Branching and Merging Collaborative Development • Pull Requests • Code Reviews • Agile Workflow	Wednesday – 22 July 2026 Morning Session	Geoffrey
	Software Engineering Best Practices	Wednesday – 22 July	Livingstone

	• Clean Code • PEP 8 Standards • Documentation	2026 Afternoon Session	
MODULE 7: DEPLOYMENT AND CAPSTONE PROJECT	Project Architecture • Structuring Python Projects • Environment Management • Requirements Files Docker Fundamentals • Docker Installation • Containers and Images • Docker Commands Cloud Deployment • Deployment Concepts • Hosting Django Applications • Environment Variables	Friday – 24 July 2026 Morning Session	Geoffrey
	Dockerizing Python Applications • Dockerfile Creation • Running Containers • Docker Compose Capstone Project Finalization • System Integration • Testing and Validation	Friday – 24 July 2026 Afternoon Session	Livingstone
	Final Project Presentations and Evaluation • Team Demonstrations • Feedback and Assessment • Guidance and Report submission	Wednesday – 29 July 2026 Morning Session	Geoffrey Livingstone
		Morning (All Groups)	Friday – 31 July 2026 Geoffrey
		Afternoon(All Groups)	Friday – 31 July 2026 Livingstone

FINAL CAPSTONE PROJECT

- DataScience Project
- AI/ML Project
- Django/Flash Web Application Project
- REST API Service
- Machine Learning Prediction System

EXPECTED LEARNING OUTCOMES

By the end of the program, learners will be able to:

- Build Python applications from scratch
- Apply object-oriented programming concepts
- Develop automation scripts
- Analyze and visualize data
- Build machine learning models
- Develop web applications and APIs
- Implement testing and debugging
- Deploy applications using Docker and CI/CD pipelines
- Collaborate using GitHub workflows

Weekly Timetable

Total weekly contact hours 9.5 hours

Time	Monday	Tuesday	Wednesday	Thursday	Friday
09:00 - 11:00	Geoffrey: jeff.geoff.cis@gmail.com	----	Geoffrey:	----	Geoffrey:
14:00-16:00	Livingstone: ndigezzalivingstone2@gmail.com	----	Livingstone:	----	Livingstone:

Note-

Group Project and Supervision Allocation- This will be done in the fifth week

Group Formation

Number: Each group will consist of 4-5 members.

Online Practical Lab Activities/ Hands-On

Platform: Online classes will be conducted via Zoom.

Schedule: Classes will follow the regular timetable.

Access: The Zoom links will be the same thought out the recess program

Links:

Morning: Register in advance for this meeting:

Dr Ruth will provide the Zoom link

Afternoon: Register in advance for this meeting

Dr Ruth will provide the Zoom link

After registering, you will receive a confirmation email containing information about joining the meeting.

Guido van Rossum (Creator of Python):

"Python is an experiment in how much freedom programmers need. Too much freedom and nobody can read another's code; too little and expressiveness is endangered."